

HackDNA – Nmap Lab 102

Challenge Overview

This lab introduces practical network reconnaissance using Nmap. The objective is to identify open ports, exposed services, and potentially vulnerable network-facing applications running on the target system.

Nmap is one of the most widely used reconnaissance and enumeration tools in cybersecurity and penetration testing.

Objective

Perform network enumeration against the target machine and identify:

- Open ports
 - Running services
 - Service versions
 - Operating system indicators
 - Potential attack surface
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What is Nmap?

Nmap (Network Mapper) is an open-source utility used for:

- Host discovery
- Port scanning
- Service enumeration
- OS fingerprinting
- Vulnerability discovery
- Network inventory

It is commonly used during the reconnaissance phase of penetration testing.

Basic Host Discovery

Before scanning services, confirm the host is online.

```
nmap -sn TARGET_IP
```

Example:

```
nmap -sn 192.168.1.10
```

This performs a ping sweep without port scanning.

Basic Port Scan

Scan the most common TCP ports.

```
nmap TARGET_IP
```

Example:

```
nmap 192.168.1.10
```

Typical output:

PORT	STATE	SERVICE
22/tcp	open	ssh
80/tcp	open	http
443/tcp	open	https

Service Version Detection

Identify software versions running on open ports.

```
nmap -sV TARGET_IP
```

Example:

```
nmap -sV 192.168.1.10
```

Possible output:

```
22/tcp open  ssh OpenSSH 8.2p1 Ubuntu  
80/tcp open  http Apache httpd 2.4.41
```

Version detection is important because outdated services may contain known vulnerabilities.

Operating System Detection

Attempt operating system fingerprinting.

```
sudo nmap -O TARGET_IP
```

Example:

```
sudo nmap -O 192.168.1.10
```

Possible results:

```
OS details: Linux 5.x
```

Aggressive Scan

Combine multiple reconnaissance techniques.

```
sudo nmap -A TARGET_IP
```

This enables:

- OS detection
- Version detection
- Script scanning
- Traceroute

Example:

```
sudo nmap -A 192.168.1.10
```

Full Port Scan

By default, Nmap scans the top 1000 ports.

To scan all TCP ports:

```
nmap -p- TARGET_IP
```

Example:

```
nmap -p- 192.168.1.10
```

This can reveal hidden services running on uncommon ports.

Faster Scanning

Adjust timing templates.

```
nmap -T4 TARGET_IP
```

Timing options range from:

- T0 → Paranoid
 - T1 → Sneaky
 - T2 → Polite
 - T3 → Normal
 - T4 → Aggressive
 - T5 → Insane
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UDP Scanning

Some important services use UDP instead of TCP.

Examples include:

- DNS
- SNMP
- DHCP
- NTP

UDP scan:

```
sudo nmap -sU TARGET_IP
```

UDP scans are slower than TCP scans.

NSE Script Scanning

Nmap includes the Nmap Scripting Engine (NSE).

Basic vulnerability scan:

```
nmap --script vuln TARGET_IP
```

Example:

```
nmap --script vuln 192.168.1.10
```

NSE scripts can detect:

- Misconfigurations
 - Weak SSL/TLS
 - Anonymous FTP access
 - SMB vulnerabilities
 - Known CVEs
-

Banner Grabbing

Service banners often reveal software versions.

Nmap can retrieve banners automatically:

```
nmap -sV TARGET_IP
```

Example banner:

```
Apache/2.4.41 (Ubuntu)
```

Attackers use this information to search for public exploits.

Common Ports and Services

Port	Service
21	FTP
22	SSH
23	Telnet
25	SMTP
53	DNS
80	HTTP
110	POP3
139	NetBIOS
143	IMAP
443	HTTPS
445	SMB
3306	MySQL
3389	RDP

Reconnaissance Workflow

Typical penetration testing workflow:

1. Host discovery
2. Port scanning
3. Service enumeration

4. Version detection
5. Vulnerability identification
6. Exploitation
7. Post-exploitation

Nmap plays a central role during early-stage reconnaissance.

Security Impact

Improperly exposed services can lead to:

- Remote code execution
- Credential theft
- Information disclosure
- Lateral movement
- Privilege escalation
- Full system compromise

Attackers rely heavily on exposed network services.

Defensive Measures

Minimize Attack Surface

Disable unused services.

Implement Firewalls

Restrict unnecessary inbound traffic.

Use Network Segmentation

Separate sensitive systems from public-facing infrastructure.

Patch Vulnerabilities

Keep software updated.

Monitor Network Activity

Deploy:

- IDS/IPS
 - SIEM solutions
 - Log analysis
 - Network anomaly detection
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Detection Indicators

Administrators can detect scanning activity through:

- Multiple connection attempts
- Sequential port probing
- Abnormal SYN packets
- IDS alerts
- Firewall logs

Tools such as:

- Snort
- Suricata
- Zeek

can help identify reconnaissance attempts.

Ethical Considerations

Unauthorized network scanning may violate:

- Organizational policies
- Terms of service
- Cybercrime laws

Always obtain proper authorization before performing security testing.

Key Takeaway

Network reconnaissance is the foundation of penetration testing. Understanding exposed services allows attackers and defenders alike to assess risk and identify security weaknesses.

Nmap remains one of the most essential tools in cybersecurity operations.

Vulnerability Classification

- CWE-200: Exposure of Sensitive Information
 - CWE-284: Improper Access Control
 - OWASP A05:2021 – Security Misconfiguration
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Conclusion

Nmap enables deep visibility into network infrastructure and exposed services. Effective enumeration can uncover weak configurations, outdated software, and hidden attack surfaces.

Defenders should continuously audit exposed services and minimize unnecessary network exposure to reduce organizational risk.